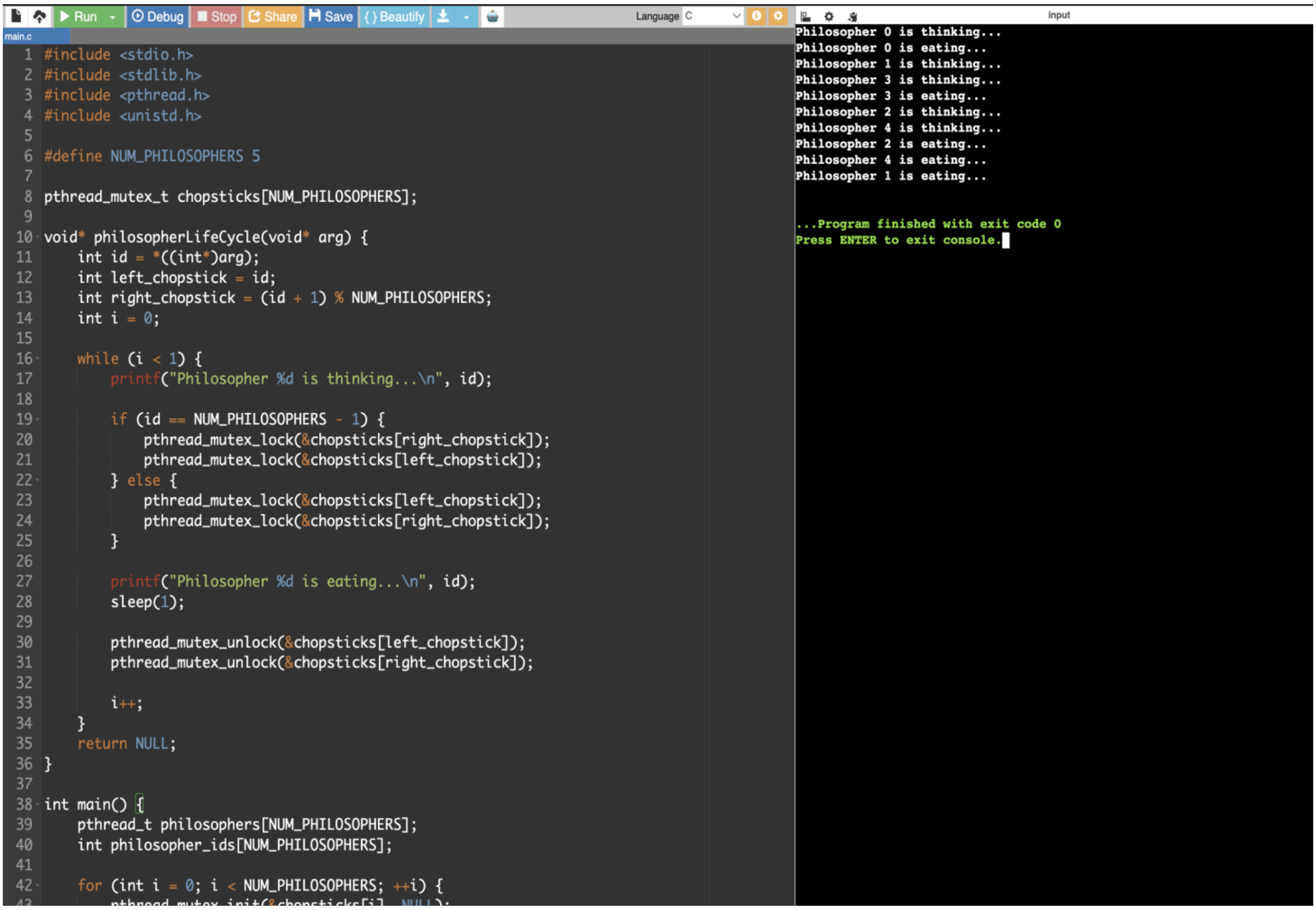
EXPERIMENT 12:

Design a C program to simulate the concept of Dining-Philosophers problem



```c
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

#define NUM_PHILOSOPHERS 5

pthread_mutex_t chopsticks[NUM_PHILOSOPHERS];

void* philosopherLifeCycle(void* arg) {
    int id = *((int*)arg);
    int left_chopstick = id;
    int right_chopstick = (id + 1) % NUM_PHILOSOPHERS;
    int i = 0;

    while (i < 1) {
        printf("Philosopher %d is thinking...\n", id);

        if (id == NUM_PHILOSOPHERS - 1) {
            pthread_mutex_lock(&chopsticks[right_chopstick]);
            pthread_mutex_lock(&chopsticks[left_chopstick]);
        } else {
            pthread_mutex_lock(&chopsticks[left_chopstick]);
            pthread_mutex_lock(&chopsticks[right_chopstick]);
        }

        printf("Philosopher %d is eating...\n", id);
        sleep(1);

        pthread_mutex_unlock(&chopsticks[left_chopstick]);
        pthread_mutex_unlock(&chopsticks[right_chopstick]);

        i++;
    }
    return NULL;
}

int main() {
    pthread_t philosophers[NUM_PHILOSOPHERS];
    int philosopher_ids[NUM_PHILOSOPHERS];

    for (int i = 0; i < NUM_PHILOSOPHERS; ++i) {
        pthread_mutex_init(&chopsticks[i], NULL);
```

```
Philosopher 0 is thinking...
Philosopher 0 is eating...
Philosopher 1 is thinking...
Philosopher 3 is thinking...
Philosopher 3 is eating...
Philosopher 2 is thinking...
Philosopher 4 is thinking...
Philosopher 2 is eating...
Philosopher 4 is eating...
Philosopher 1 is eating...

...Program finished with exit code 0
Press ENTER to exit console.
```